Theory

ISPhO

A2-1

Korean (Korea (Republic of))

폴리인의 진동 구조

Part A. 무한한 탄소 원자 사슬 (6.5점)

A.1 (0.8pt)

$E =$

A.2 (0.8pt)

$m\ddot{u}_n =$

$m\ddot{v}_n =$

A.3 (0.9pt)

$a =$ $b =$

$c =$ $d =$

A.4 (1.0pt)

$\omega_1/\omega_0 =$

$\omega_2/\omega_0 =$

A.5 (1.0pt)

$\omega_1/\omega_0 \approx$

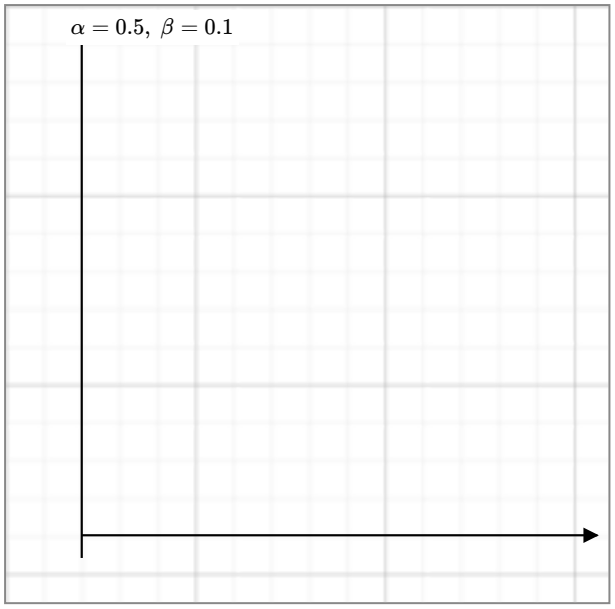
$\omega_2/\omega_0 \approx$

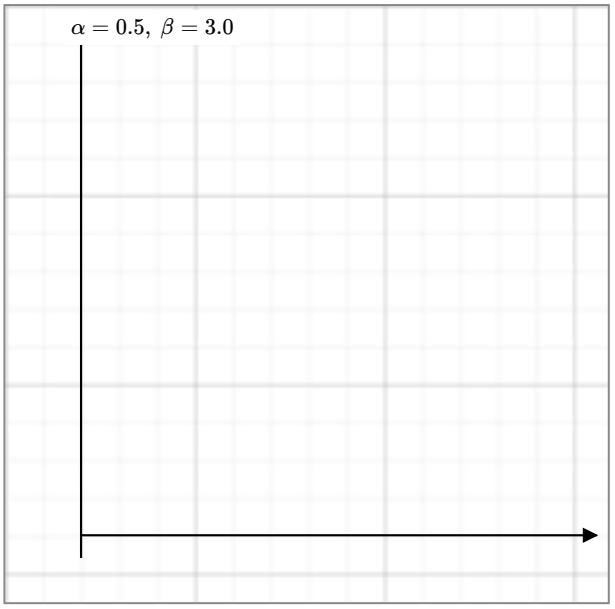
A.6 (1.0pt)

$\omega_1/\omega_0 \approx$

$\omega_2/\omega_0 \approx$

A.7 (1.0pt)
그래프:





Part B. 폴리인의 분광학 (3.5점)

B.1 (0.8pt)

식:

B.2 (0.7pt)

$q_l =$

B.3 (2.0pt)

단일 결합:

삼중 결합: